

- 2 Cancer of the blood, including leukaemia and lymphoma, can be caused by mutations of stem cells in the bone marrow.

A long-term study into the effects of radiation on the frequency of blood cancer was carried out on two groups of people: group 1 and group 2. These people were all born to mothers exposed to nuclear radiation during pregnancy.

Table 2.1 summarises information about the two groups of people included in this study.

Table 2.1

	group 1	group 2
when born	between 1948 and 1988	between 1950 and 1961
how the mothers were exposed to radiation	working in a nuclear power plant and living in the town next to the nuclear power plant	living next to a river contaminated by nuclear wastes from an accident at the same nuclear power plant
time when mothers were exposed to radiation	any time between January 1948 and December 1982	any time between January 1950 and December 1960
method of determining radiation exposure of mothers	using badges worn by workers at the nuclear power plant to record their exposure to radiation	from external radiation levels measured in the area
individuals for whom blood cancer data were collected	people who continued to live in the same town as the nuclear power plant	people who continued to live in the area where they were born
when blood cancer data were collected	January 1948 until December 2009	January 1953 until December 2009

Until 2005, the data sources used for all of this information were paper based and obtained from hospitals, clinics and medical records.

After 2005, data were collected electronically from databases at cancer clinics and from online death certificates.

(a) Table 2.2 shows some of the results from this study.

Table 2.2

	group 1	group 2	group 1 and group 2 combined
number of people in the group studied	8 466	11 070	19 536
male	4 361	5 588	9 949
female	4 105	5 482	9 587
outcomes up to December 31 2009			
number of people not developing any cancer who were still alive	4 053	5 648	9 701
number of people not developing any cancer who had died	898	1 864	2 762
number of people developing any cancer	220	288	508
number of deaths from any cancer	103	145	248
number of people developing blood cancer	32	26	58
number of deaths due to blood cancer	21	15	36
number of people where outcome not known	3 295	3 270	6 565

- (i) In group 1, the proportion of people who were known to develop blood cancer out of all the people who were known to develop any cancer was 0.145.

Calculate for group 2 the proportion of people who were known to develop blood cancer out of all the people who were known to develop any cancer.

Give your answer to three decimal places.

..... [1]

- (ii) It is possible to carry out a chi-squared test on the data in Table 2.2 to test whether there is a difference in the probability of individuals in group 1 and group 2 developing blood cancer.

State **one** reason why the chi-squared test can be used with these data.

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 [1]

- (b) The data were analysed to assess how the number of people who developed blood cancer was affected by their mothers' exposure to radiation during pregnancy.

Fig. 2.1 shows the results of this analysis for the combined data from group 1 and group 2. Each plotted number includes all those people whose mothers' exposure to radiation during pregnancy was below, or up to, the exposure to radiation shown.

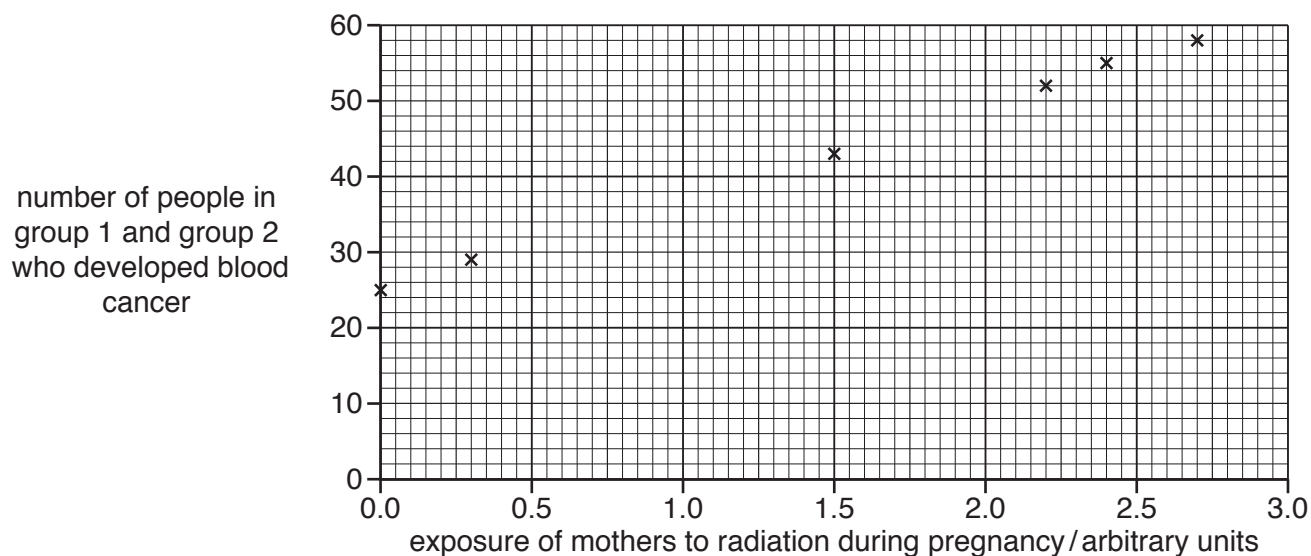


Fig. 2.1

- (i) Use Fig. 2.1 to describe the relationship between the number of people who developed blood cancer and their mothers' exposure to radiation during pregnancy.

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- (ii) Suggest an explanation for the relationship shown between the number of people who developed blood cancer and their mothers' exposure to radiation during pregnancy.

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- (c) Evaluate the validity of the results of this study with reference to all the information provided.

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- (d) Plant scientists were interested in the effect of radiation on the germination of seeds.

They exposed seeds to the same intensity of radiation for different lengths of time and measured the proportion of seeds that germinated.

Suggest **three** variables, other than intensity of radiation, that would need to be standardised in an investigation of this type.

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..... [3]

[Total: 12]

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